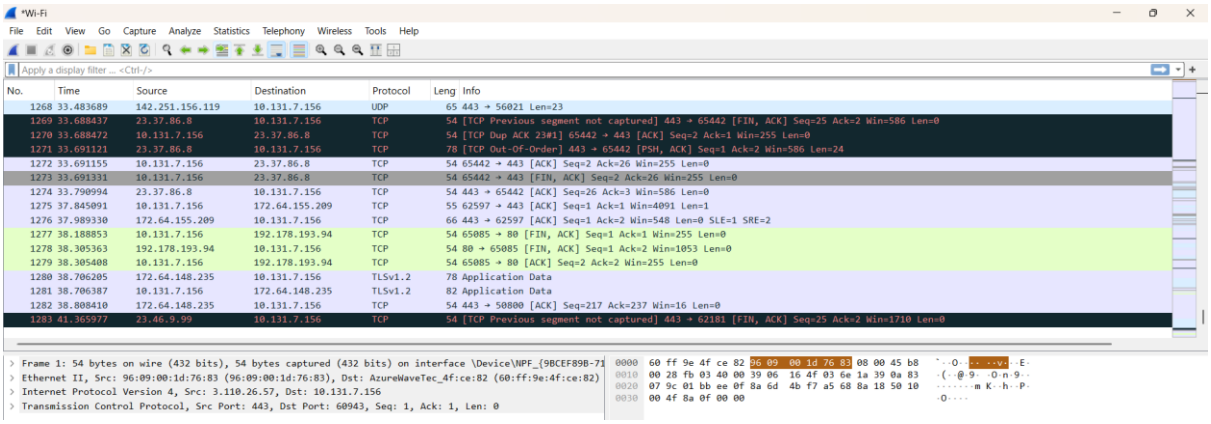
3: Using Advanced ToolsTheory :Introduction to tools like Wireshark, Nmap.Practical :Use Wireshark to capture and analyze network traffic.Task: Document key findings from a packet capture.

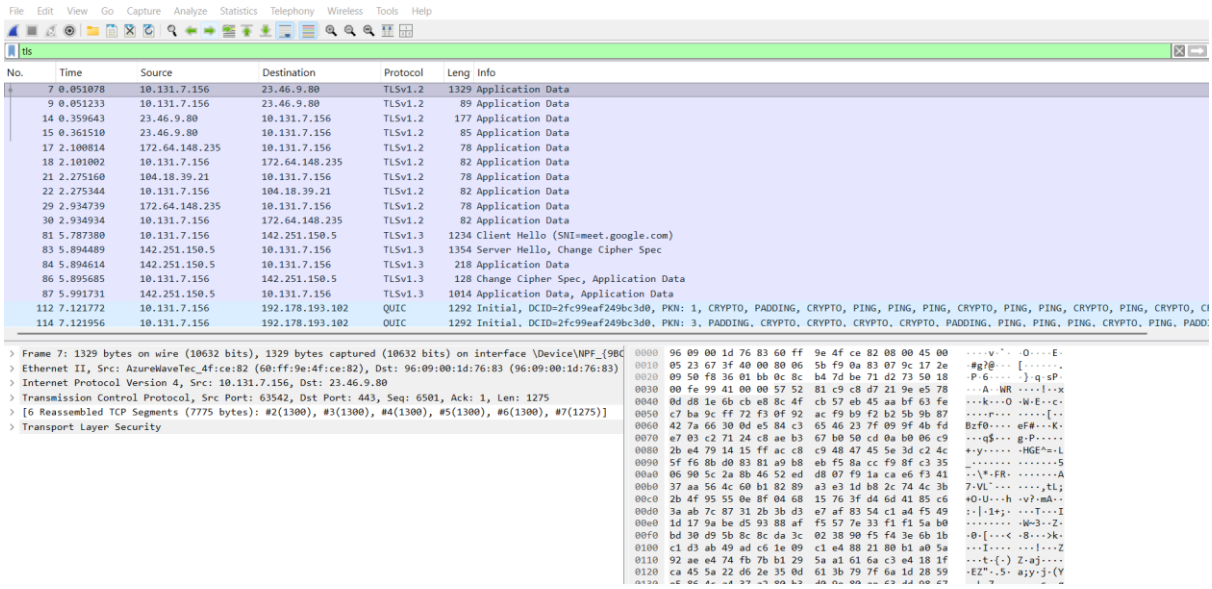
Task 1.

Install Wireshark on your computer and capture 2 minutes of network traffic while you browse Instagram or YouTube. Save the capture file as insta_traffic.pcap.



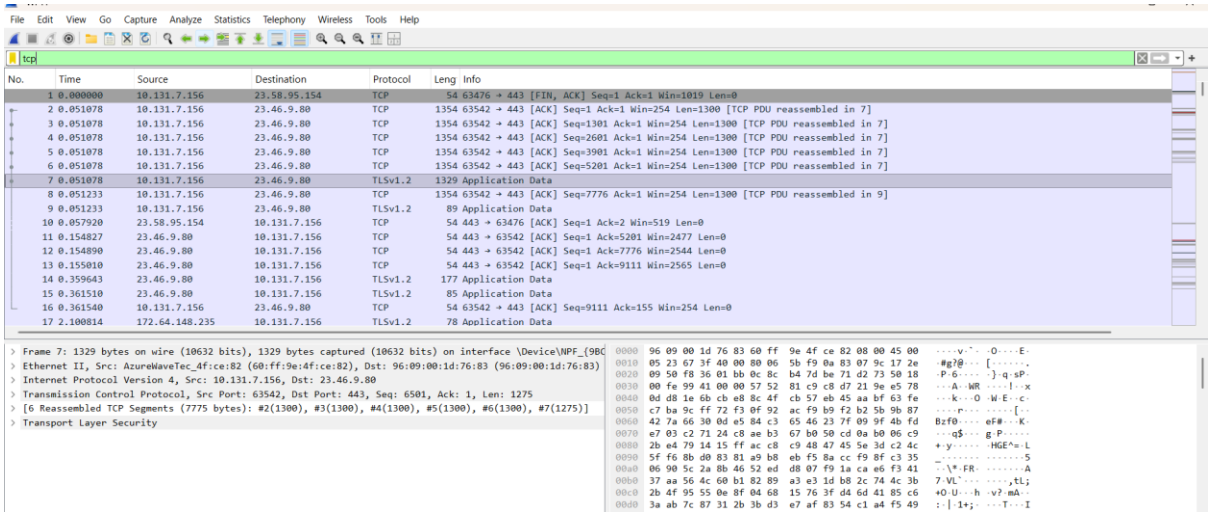
Task 2.

Use Wireshark to filter and list all HTTP or HTTPS requests made by your browser during the capture. Write down the domains you see most frequently.



Task 3.

Open your insta_traffic.pcap in Wireshark and find a TCP packet. Document the source IP, destination IP, and TCP port numbers for that packet.



Task 4.

Run an Nmap scan on your own machine (localhost) to detect open ports. Save the command you used and list the open ports found.

Hint: Use 'nmap 127.0.0.1' in your terminal.

Check the installation

nmap --version

Scan your own machine

nmap 127.0.0.1

If the command is not recognized, reopen Command Prompt after installation or run Nmap using its full executable path.

Record the results

Nmap displays the ports it detects as open, closed, or filtered. Only record ports marked open.

Example of the output format:

PORT STATE SERVICE

135/tcp open msrpc

445/tcp open microsoft-ds

This is an illustrative example; your results may be different. A localhost scan checks services listening on your own computer, not every device on your network.

Your assignment record should contain:

- Command used: nmap 127.0.0.1
- Open TCP ports: enter those shown as open
- Service names: enter the names Nmap reports

Task 5.

Capture network traffic while using a food delivery app (like Zomato or Swiggy) for 1 minute. In Wireshark, identify at least one DNS request and document the domain name being resolved.

Hint: Use the 'dns' filter in Wireshark to find DNS packets.

